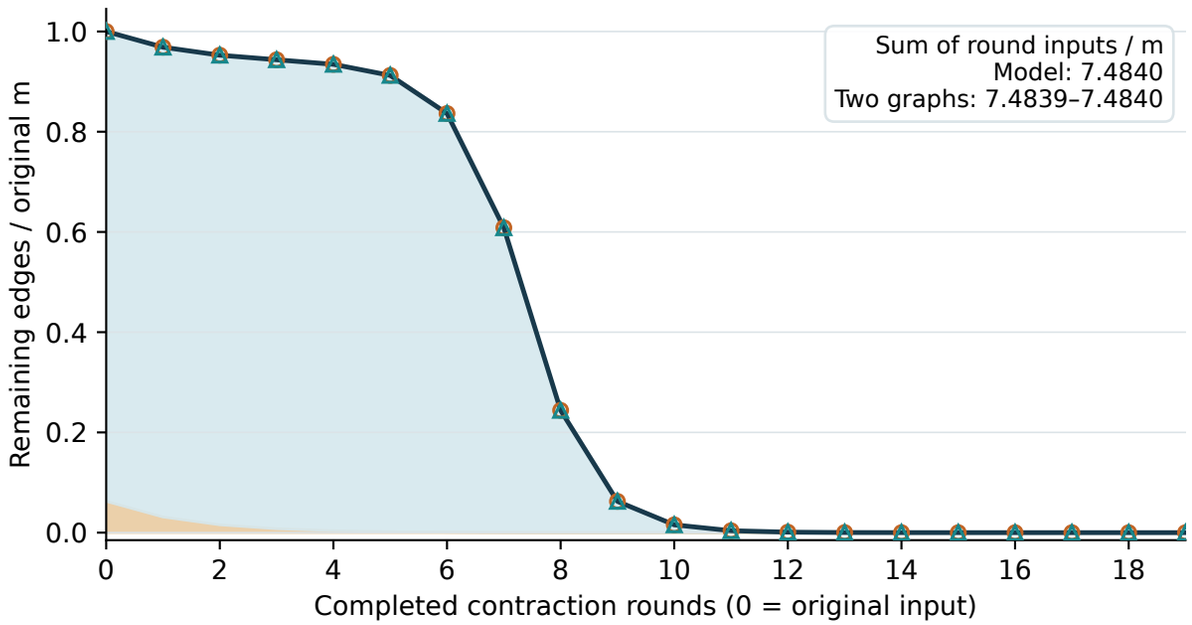
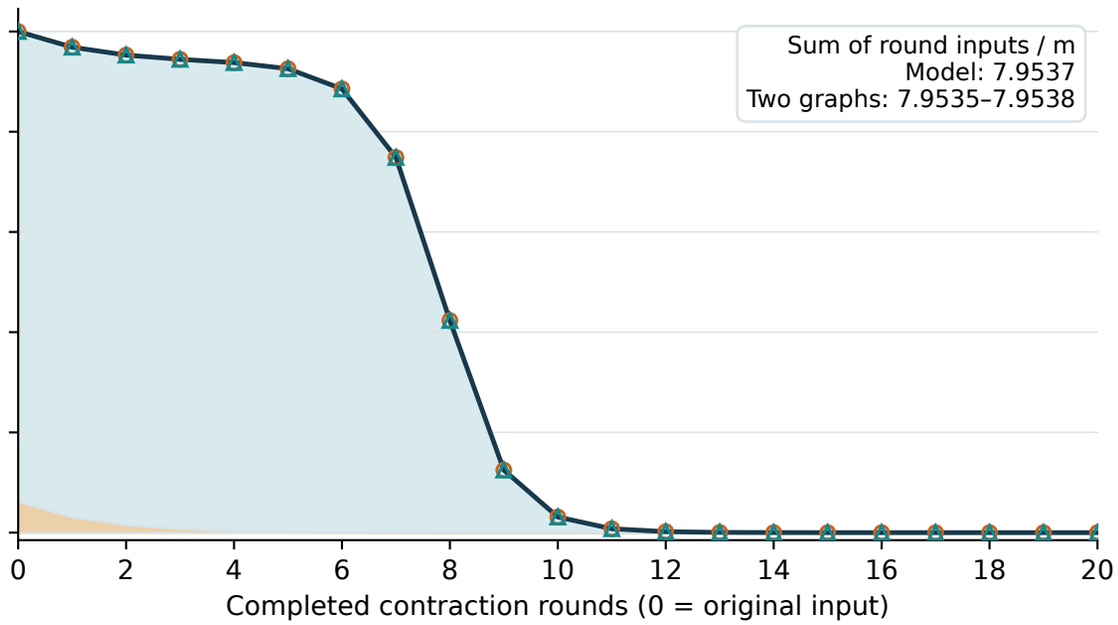
Edges retained by continued production contraction

m = 16n · Zero-intrusion engineered hierarchies · A theoretical expectation and two observed graph draws

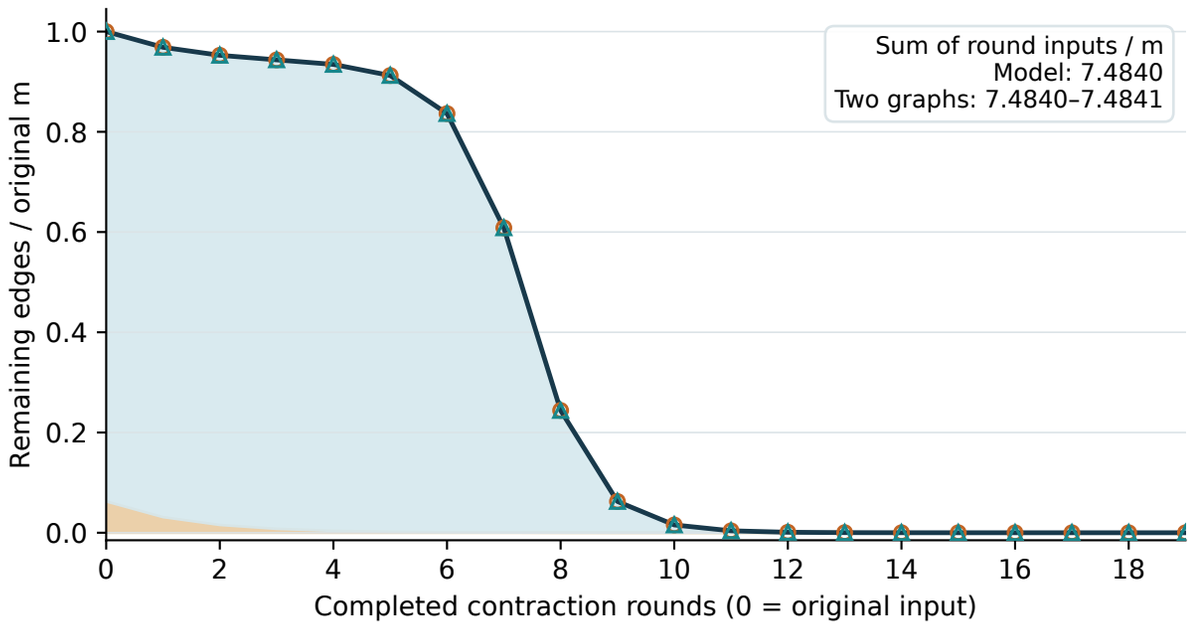
Hierarchical path · n = 524,288



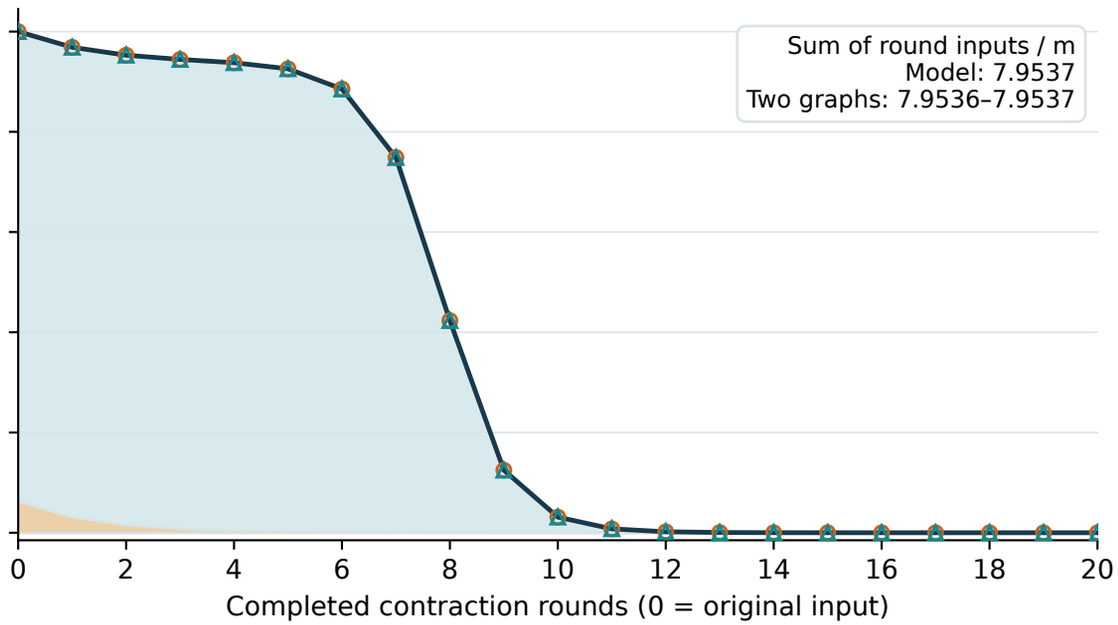
Hierarchical path · n = 1,048,576



Hierarchical tree · n = 524,288



Hierarchical tree · n = 1,048,576



Guaranteed backbone component pairs
Occupied shortcut-only pairs: expectation
Conditional expected total
Observed graph seed 17
Observed graph seed 42

Shading decomposes expected occupied component pairs; it is not a confidence band. These are edge counts, not time predictions.